

2(a)	work done per unit mass	B1
	(work done in) moving mass from infinity	B1
2(b)(i)	(gravitational) fields from the Earth and Moon are in opposite directions	B1
	(resultant is zero where gravitational) fields are equal (in magnitude)	B1
2(b)(ii)	$g \propto M / r^2$	C1
	$5.98 \times 10^{24} / x^2 = 7.35 \times 10^{22} / (3.84 \times 10^8 - x)^2$	A1
	leading to $x = 3.5 \times 10^8$ (m)	
2(b)(iii)	ϕ (Earth) = $(-6.67 \times 10^{-11} \times (5.98 \times 10^{24} / 3.5 \times 10^8))$ and ϕ (Moon) = $(-6.67 \times 10^{-11} \times (7.35 \times 10^{22} / 0.38 \times 10^8))$	C1
	$\phi = (-6.67 \times 10^{-11} \times [(5.98 \times 10^{24} / 3.5 \times 10^8) + (7.35 \times 10^{22} / 0.38 \times 10^8)])$	C1
	$= -1.3 \times 10^6 \text{ J kg}^{-1}$	A1